Facilities and Other Resources

ONR FY2027 YIP, NOFO N0001426SF004. PI: Sterling G. Baird, Brigham Young University

PI laboratory, BYU Department of Mechanical Engineering. A powder metallurgy laboratory [TODO Sterling: square footage and room] holds an AMAZEMET rePOWDER ultrasonic atomizer with an induction crucible module, which makes aluminum alloy powder from elemental and master-alloy feedstock weighed into machined crucibles of high-purity aluminum, four compositions per week. The room has a canopy hood, a fume hood, an industrial dehumidifier, dry powder storage, and electrical, gas, compressed air, and chilled water services, installed with BYU Environmental Health and Safety. The laboratory connects directly to the BYU Microscopy Center by a door with no public access.

BYU Microscopy Center. Scanning electron microscopes with energy-dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD) (Apreo C, Helios 600 with a focused ion beam, and Verios), and a Tecnai TF-20 transmission electron microscope. Instruments are shared and have high availability.

Other BYU facilities. Metallographic preparation (PX500 automatic and PX300 manual polishers, vibratory polishers, fume hood for etching, and an electropolisher); a Keyence VHX-7000 digital optical microscope; the BYU X-ray Diffraction Facility; the BYU Spectroscopic Facility (Agilent 8900 ICP-MS/MS for composition, and XPS); and the BYU Prototyping Lab (CNC machining for crucibles and test specimens). Tensile tests use [TODO Sterling: name the BYU load frame or test service].

Computing. The BYU Office of Research Computing operates a shared cluster of about 28,000 CPU cores and 367 GPUs on 616 nodes (`https://rc.byu.edu`) [TODO Sterling: update the counts if they have changed since April 2026], available at no direct cost through fair-share scheduling. CALPHAD calculations use Thermo-Calc [TODO Sterling: confirm the BYU license and the aluminum databases] and pycalphad.

Laser powder bed fusion. Builds are bought on a purchase order at $800 per build from a University of Utah user facility [TODO Sterling: facility name, printer model, and build envelope; the 2026 DOE proposal named an Aconity MIDI with a 170 mm build plate and a 200 to 1000 W fiber laser].

Corrosion testing. The electrochemical screen uses a potentiostat requested in Year 1 (see Field 11). Neutral salt-spray exposures are bought from an outside test laboratory [TODO Sterling: name the laboratory once quoted].

Government-owned facilities. None are required [TODO Sterling: revise if a Navy laboratory agrees to run stress corrosion tests on final candidates].